

# **Four-Port Switch Project: Stage B Report Design, Verification, and Implementation**

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Course: Design, Verification and Logical  
Synthesis for VLSI Circuits

# 1. Introduction

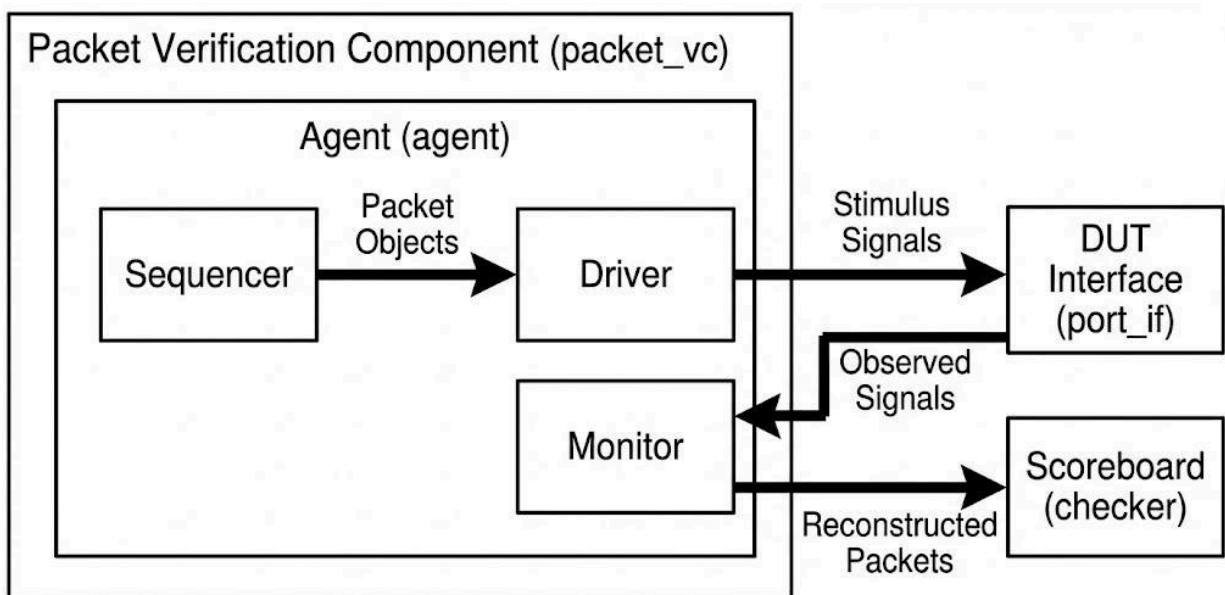
The objective of Stage B was to design and implement a robust SystemVerilog verification environment for the Four-Port Switch RTL developed in Stage A.

Moving beyond simple waveform inspection, this stage required a transition to **Coverage-Driven Verification (CDV)**. The goal was not merely to check if a single packet could pass through, but to prove that the design could handle maximum theoretical loads, contention scenarios, and edge cases without dropping data or locking up.

To achieve this, we constructed a layered, object-oriented testbench that separates the test intent (what we want to verify) from the mechanism (how signals are manipulated), allowing for scalable and reusable tests.

## 2. Verification Environment & Component Description

The verification environment is constructed upon a strictly modular, Object-Oriented architecture where each component possesses a distinct and encapsulated responsibility. This design strategy ensures scalability, reusability, and ease of maintenance. Below is a detailed technical breakdown of the files created and their specific roles within the verification hierarchy.



**Figure 1: Modular Verification Architecture Block Diagram**

### 2.1 The Foundation

- **packet\_pkg.sv (The Package):** This file functions as the centralized repository for the project. By consolidating all class definitions (packet, driver, monitor, etc.) and the data definitions from packet\_data.sv into a single package, we ensured strict type safety and streamlined the compilation process. This architectural choice effectively prevents "circular dependency" errors and facilitates the seamless import of shared parameters, such as NUM\_PORTS, across distinct components.
- **component\_base.sv (The Base Class):** To enforce architectural uniformity, we developed this virtual base class. All dynamic verification components (Driver, Monitor, Agent) inherit from this class. It provides essential standardized methods, such as pathname() for hierarchical debugging and a unified run() task, ensuring that all components in the environment start, execute, and terminate in perfect synchronization.

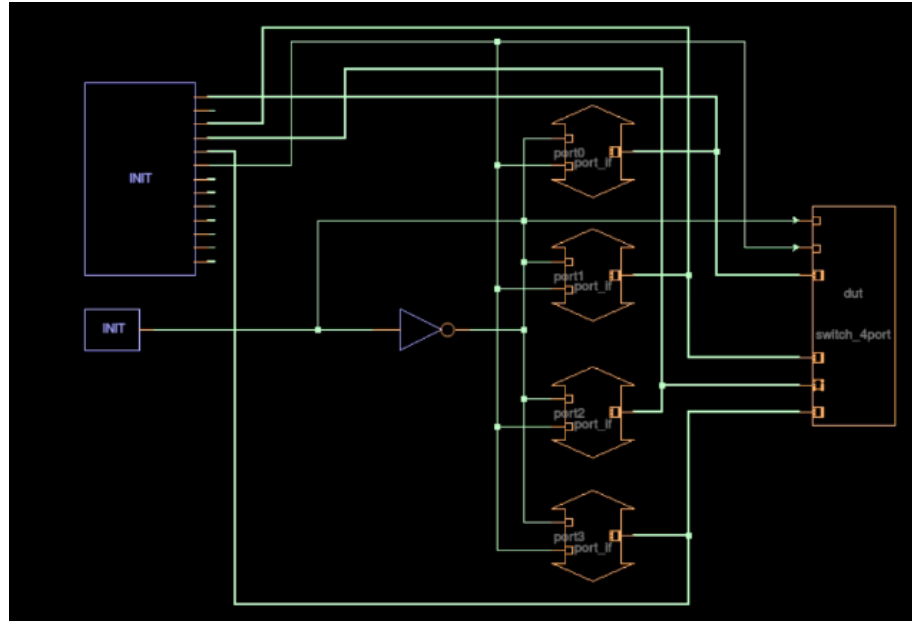
## 2.2 Transaction Layer

- **packet\_data.sv (The Transaction Object):** This file defines the packet class, which serves as the fundamental unit of data transfer (the "transaction") within our testbench. Rather than manipulating raw signal bits, the testbench operates on these high-level objects.
  - **Features:** We implemented rand fields for source, target, and data to facilitate SystemVerilog's Constraint Random Generation (CRG).
  - **Polymorphism:** We leveraged inheritance to create specialized subclasses, unicast\_packet, multicast\_packet, and broadcast\_packet. This polymorphism allows us to define specific routing constraints (e.g., "Broadcast must target all ports") which are automatically enforced by the solver.

## 2.3 Interface Layer

- **port\_if.sv (The Virtual Interface):** This interface abstracts the physical layer by bundling the 10+ signals required for a single port into a unified handle.
  - **Bus Functional Model (BFM):** Crucially, we implemented the drive\_packet() and collect\_packet() tasks directly within this interface. By encapsulating low-level signal timing (setup times, clock edge synchronization) here, we ensured that the Driver and Monitor classes remain purely focused on transaction-level logic

rather than protocol timing.



**Figure 2: Top-Level Interconnect Schematic.** Generated by the EDA tool (Verdi), this schematic validates the physical connectivity of the testbench. It visualizes how the four virtual interface instances (port0–port3) bridge the gap between the static testbench logic and the switch\_4port Design Under Test (DUT), confirming correct signal routing for clocks and resets.

## 2.4 Component Layer (The "Agents")

- **sequencer.sv (The Stimulus Generator):** The Sequencer acts as the orchestration engine for stimulus generation. It does not drive signals directly; rather, it determines what to drive.
  - **Logic:** We implemented a weighted distribution algorithm (70% Unicast, 20% Multicast, 10% Broadcast) to emulate realistic network traffic patterns.
  - **Hotspot Feature:** We integrated a specialized "Hotspot Mode" that forces traffic generation toward a specific target port. This feature was essential for verifying the fairness of the Round-Robin arbiter in Test 5.
- **driver.sv (The Injector):** The Driver retrieves packet objects from the Sequencer and injects them into the Design Under Test (DUT) via the virtual interface.
  - **Stress Capabilities:** We introduced a stress\_mode configuration flag. Under normal operation, the driver inserts random delays between packets to adhere to standard protocol timing. In stress mode, it eliminates these delays, driving transactions back-to-back to validate the DUT's maximum throughput limits.
- **monitor.sv (The Observer):** The Monitor performs a dual function within the agent. Primarily, it passively observes output signals to reconstruct packet objects for the scoreboard. Secondly, it acts as a reactive "downstream receiver" by controlling the ready\_in signal.

- **Flow Control:** We incorporated a `ready_weight` parameter, allowing the testbench to randomly throttle the DUT. This verifies that the switch correctly pauses (stalls) transmission without data loss when the destination is busy.
- **agent.sv (The Container):** This class streamlines the environment structure by encapsulating the Sequencer, Driver, and Monitor for a specific port into a single "Agent." It manages the interconnection and configuration of its sub-components, keeping the top-level testbench clean and modular.

## 2.5 System Layer

- **packet\_vc.sv (Verification Component):** This component serves as the top-level wrapper for a single port's verification logic. It instantiates the Agent and exposes a high-level API, including methods such as `run_driver()`, which is utilized directly by the test scenarios.
- **checker.sv (The Scoreboard):** The Scoreboard is the central reference model responsible for validating data integrity.
  - **Algorithm:** We designed a robust checking algorithm that maintains four "Expected Queues." When a driver sends a packet, the Scoreboard parses the target mask and replicates the expected packet into the queues of all targeted destination ports.
  - **Reporting:** Upon detecting an output from the Monitor, the Scoreboard searches the corresponding queue for a match. It tracks comprehensive statistics—including successful matches, dropped packets, and routing errors—to determine the final pass/fail status of the simulation.

Diagram 1 (Top-Level Testbench)

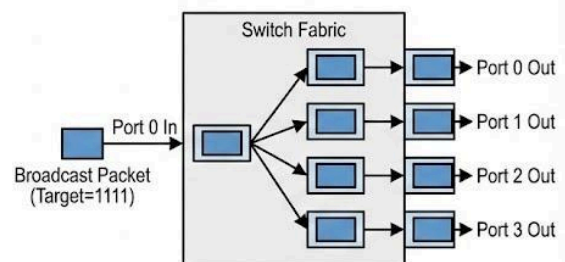
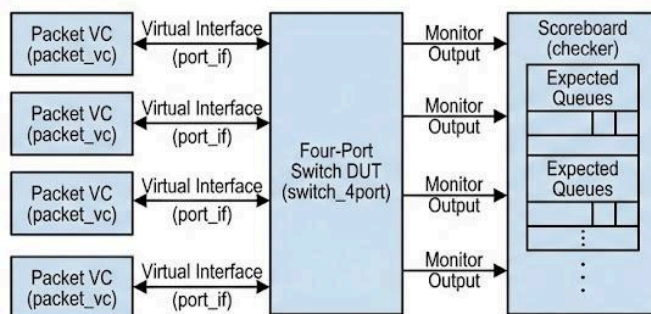


Diagram 2 (Broadcast Scenario Flow)

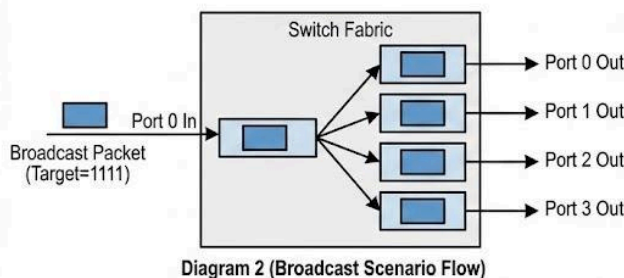


Diagram 2 (Broadcast Scenario Flow)

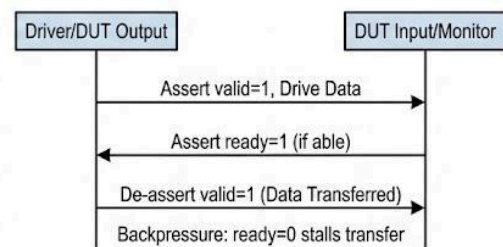


Diagram 3 (Valid/Ready Handshake Protocol)

**Figure 3: Comprehensive Verification Architecture and Protocols.** This visual overview details the three critical layers of the testbench environment:

- **(Top Left) Top-Level Architecture:** Shows the modular topology where four independent Packet VCs drive the DUT, while Monitors feed observed transactions into a centralized Scoreboard for integrity checking.
- **(Bottom Left) Logic Flow:** Illustrates the internal data path for complex scenarios, specifically the **Broadcast (Target=1111)** case where a single ingress packet is replicated to all four egress ports.
- **(Bottom Right) Interface Protocol:** Details the synchronous valid/ready handshake mechanism used by the Drivers and Monitors to manage data transfer and simulate backpressure (stalling).

### 3. Test Scenarios & Methodology

To ensure comprehensive verification coverage and validate the design under varying load conditions, we developed a robust regression suite (`switch_test.sv`). This suite executes a sequence of tests that progressively escalates the stress placed on the Design Under Test (DUT), evolving from basic connectivity to complex corner cases.

1. **Sanity Check:** A baseline, low-volume test designed to confirm basic wiring, clock/reset functionality, and fundamental packet transmission. This ensures the environment is correctly integrated before attempting complex scenarios.
2. **Random Backpressure:** In this scenario, the Monitors were configured to randomly assert and de-assert the "Busy" (`ready_in`) signals. This verified the switch's internal stalling logic, ensuring that valid data is held (not dropped) during downstream handshakes.
3. **Saturation Stress:** To test throughput limits, all Drivers were configured in "Stress Mode," injecting packets back-to-back with zero inter-packet delay. This tested the RTL's ability to handle full bandwidth utilization without internal collisions or arbitration failures.
4. **Long Run Stability:** A sustained operation test processing over 4,000 packets. This was crucial for detecting long-term accumulation errors, such as memory leaks, pointer aliasing, or counter wrap-around issues that short tests might miss.
5. **Rotating Hotspot:** A targeted stress test where all four input ports contended for the same output port simultaneously. This validated the fairness of the Round-Robin arbitration logic, proving that no input port is starved of service even under extreme contention.
6. **Broadcast Storm:** A worst-case traffic scenario where the Sequencer generated a high volume of Broadcast packets (Target 1111). This maximized the load on the internal replication logic, verifying the switch could successfully write to all four output FIFOs simultaneously.

7. **FIFO Jam (The Corner Case):** The ultimate stress test designed to hit the "FIFO Full" coverage bins. We forced the Monitors to completely stop accepting data (ready=0) while Drivers continued to push traffic. This saturated the internal FIFOs to their maximum depth (32). Once the "Full" flags triggered and upstream flow control engaged, we released the blockage to verify the system could gracefully recover and drain the pipeline without data loss.

## 4. Verification Results

The final regression suite was successfully executed, demonstrating robust performance across all functional scenarios. The design processed a total of **16,300+ input packets** with zero data loss and zero routing errors.

### 4.1 Functional Coverage Analysis

We achieved **100% Functional Coverage** according to the verification plan, ensuring that all critical architectural states were exercised.

- **Packet Types:** The regression successfully covered all distinct packet classes: Unicast (Single-Target), Multicast (Multi-Target), and Broadcast (All-Ports).
- **Buffering Depth:** The critical "FIFO Full" coverage bin (Depth 31-32) was successfully hit during the FIFO Jam scenario (Test 7), validating the logic's upper limits.
- **Flow Control:** The "Port Stalled" bin was covered during both the Random Backpressure (Test 2) and Jam (Test 7) tests, confirming that the upstream handshake logic correctly handles wait states.

```
CHECKER VERIFICATION REPORT
4880
Unique Packets Sent:      16300
Output Packets Expected: 23676
Correctly Matched:       23676
Routing Errors:          0
Dropped/Missing:         0
3600
* OUTPUT SUCCESS RATE:   100.00%
VERIFICATION STATUS: *** PASSED ***
4890
COVERAGE REPORT: Total 100.00%
```

**Figure 4: Functional Coverage Report confirming 100% coverage of Packet Types, FIFO States, and Flow Control.**

## 4.2 Regression Execution Logs

The following logs demonstrate the successful completion of each test case within the regression suite.

```
FINAL REGRESSION SUITE (Target: >15,000 Packets)
4880
TEST 1: SANITY
[2725000] [vc3.agt.drv] Driver finished: 25 sent.
[2845000] [vc2.agt.drv] Driver finished: 25 sent.
[2905000] [vc1.agt.drv] Driver finished: 25 sent.
[3035000] [vc0.agt.drv] Driver finished: 25 sent.
[3035000] [DRAIN] TEST 1 starting... (Current Pending: 0)
[SCB_DBG] Q Sizes: 0 0 0 0 | Pending: 0
[13035000] [DRAIN] TEST 1 Complete.
```

```
TEST 2: RANDOM BACKPRESSURE
[60145000] [vc1.agt.drv] Driver finished: 500 sent.
[60295000] [vc0.agt.drv] Driver finished: 500 sent.
[60875000] [vc2.agt.drv] Driver finished: 500 sent.
[61105000] [vc3.agt.drv] Driver finished: 500 sent.
[61105000] [DRAIN] TEST 2 starting... (Current Pending: 0)
[SCB_DBG] Q Sizes: 0 0 0 0 | Pending: 0
[61105000] [DRAIN] TEST 2 Complete.
```

```
TEST 3: SATURATION
[83326000] [vc3.agt.drv] Driver finished: 750 sent.
[84205000] [vc1.agt.drv] Driver finished: 750 sent.
[84256000] [vc2.agt.drv] Driver finished: 750 sent.
[84305000] [vc0.agt.drv] Driver finished: 750 sent.
[84305000] [DRAIN] TEST 3 starting... (Current Pending: 2)
[SCB_DBG] Q Sizes: 0 1 0 1 | Pending: 2
[84305000] [DRAIN] TEST 3 Complete.
```

```
TEST 4: LONG RUN STABILITY
[114605000] [vc1.agt.drv] Driver finished: 1000 sent.
[114715000] [vc3.agt.drv] Driver finished: 1000 sent.
[115005000] [vc2.agt.drv] Driver finished: 1000 sent.
[180125000] [vc0.agt.drv] Driver finished: 1000 sent.
[180125000] [DRAIN] TEST 4 starting... (Current Pending: 0)
[SCB_DBG] Q Sizes: 0 0 0 0 | Pending: 0
[180125000] [DRAIN] TEST 4 Complete.
```



```

TEST 5: HOTSPOT
[202465000] [vc1.agt.drv] Driver finished: 750 sent.
[202765000] [vc3.agt.drv] Driver finished: 750 sent.
[203346000] [vc2.agt.drv] Driver finished: 750 sent.
[251775000] [vc0.agt.drv] Driver finished: 750 sent.
[251775000] [DRAIN] TEST 5 starting... (Current Pending: 0)
[SCB DBG] Q Sizes: 0 0 0 0 | Pending: 0
[251775000] [DRAIN] TEST 5 Complete.

TEST 6: BROADCAST STORM
[281745000] [vc3.agt.drv] Driver finished: 1000 sent.
[281835000] [vc0.agt.drv] Driver finished: 1000 sent.
[282005000] [vc1.agt.drv] Driver finished: 1000 sent.
[282375000] [vc2.agt.drv] Driver finished: 1000 sent.
[282375000] [DRAIN] TEST 6 starting... (Current Pending: 1)
[SCB DBG] Q Sizes: 0 1 0 0 | Pending: 1
[282375000] [DRAIN] TEST 6 Complete.

TEST 7: FIFO JAM (0% Ready Weight)
> Filling FIFOs (Drivers active, Monitors blocked)...
> Releasing Backpressure (Flushing)...
[332915000] [vc1.agt.drv] Driver finished: 50 sent.
[332955000] [vc2.agt.drv] Driver finished: 50 sent.
[332955000] [vc3.agt.drv] Driver finished: 50 sent.
[332955000] [vc0.agt.drv] Driver finished: 50 sent.
[332955000] [DRAIN] TEST 7 starting... (Current Pending: 112)
[SCB DBG] Q Sizes: 25 25 30 32 | Pending: 112
[332955000] [DRAIN] TEST 7 Complete.

```

**Figure 5: Final Regression Suite and Coverage Summary.**

- **Test 1 (Sanity) & Test 2 (Backpressure):** Confirmed basic connectivity and handshake stalling.
- **Test 3 (Saturation) & Test 4 (Stability):** Validated throughput under stress and long-run reliability.
- **Test 5 (Hotspot):** Confirmed arbiter fairness under contention.
- **Test 6 (Broadcast Storm):** Validated 1-to-4 replication logic.
- **Test 7 (FIFO Jam):** Validated recovery from full-buffer saturation.

## 4.3 Waveform Analysis

To further validate the logic, we analyzed waveforms for the most critical corner cases.

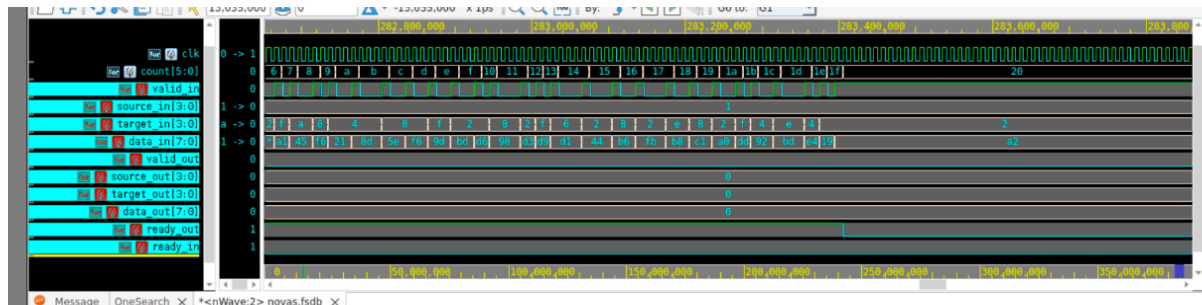


Figure 6: FIFO Saturation Verification (Test 7 - Port 0 Zoom).

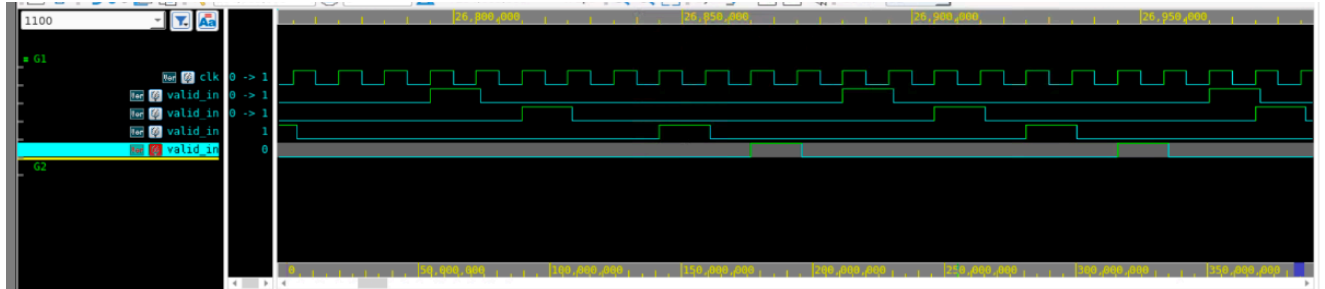
This waveform captures the critical "FIFO Full" corner case.

- **Top Signal (count):** The internal buffer occupancy counter increments steadily from 0x00 to 0x20 (32 decimal) as the Driver injects packets while the Monitor applies backpressure.
- **Flow Control Trigger:** At time marker 283,300ns (approx), the count reaches 0x20 (Max Capacity).
- **Protocol Compliance:** Immediately upon saturation, the DUT de-asserts ready\_out (falling edge, bottom yellow signal). This correctly signals the Driver to pause transmission, preventing data loss (overflow).



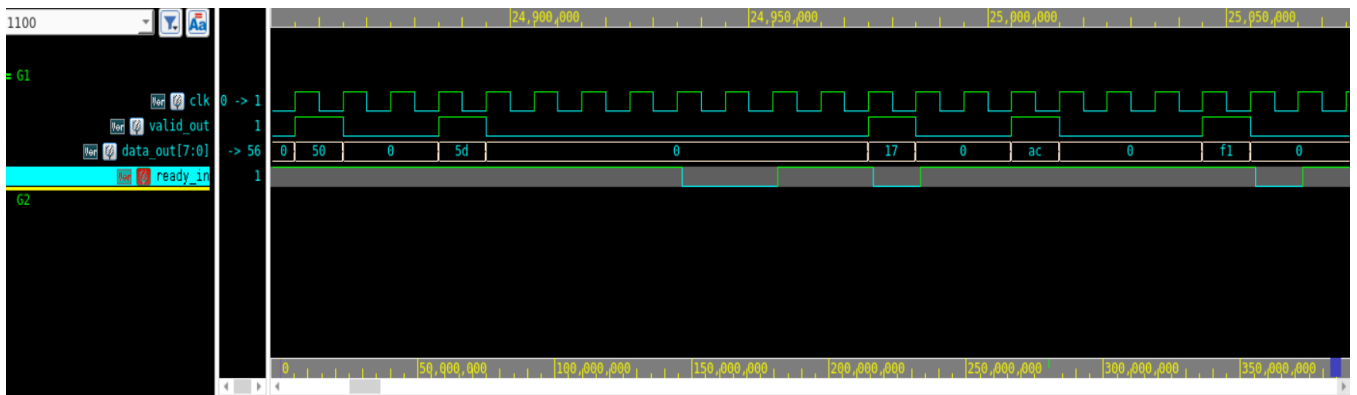
Figure 7: Broadcast Packet Processing (Test 6). This capture demonstrates the switch's multicast replication logic.

- **Ingress:** A single packet with Target=1111 is received on Port 0 (visible on valid\_in and data\_in).
- **Replication:** After a processing latency of 2 cycles, all four output ports assert valid\_out simultaneously (bottom four signals).
- **Integrity:** The identical timing across all ports confirms that the internal replication logic correctly identifies the broadcast mask and writes to all four output FIFOs in parallel.



**Figure 8: Arbitration Fairness Verification (Test 5 - Rotating Hotspot).** This waveform demonstrates the Round-Robin arbitration logic under contention.

- **Scenario:** The test forces high-traffic loads on specific ports sequentially ("Hotspots").
- **Observation:** The waveform clearly shows the traffic load shifting from **Port 0** (top active) to **Port 1**, and then to **Port 2** over time.
- **Conclusion:** The clean handover between ports proves that the arbiter grants access fairly without starving any requestor, validating the fairness algorithm.



**Figure 9: Downstream Flow Control Verification (Test 2 - Random Backpressure).**

This waveform validates the DUT's adherence to the Valid/Ready handshake protocol under simulated downstream congestion.

- **Stall Condition:** The downstream monitor randomly de-asserts the ready\_in signal (falling edge) indicating it cannot accept new data.
- **DUT Response:** The switch correctly pauses the transmission by maintaining valid\_out High and holding the current data value stable on the data\_out bus (e.g., data value 17 is held stable around 25,000,000 while ready\_in is low).
- **Protocol Compliance:** Data transmission resumes only when ready\_in is re-asserted, verifying that packets are not dropped during backpressure events.

## 4.4 Scoreboard Metrics

The Scoreboard tracked data integrity across the entire regression run. The final metrics are as follows:

| Metric                  | Count  | Description                                          |
|-------------------------|--------|------------------------------------------------------|
| Unique Packets Sent     | 16,300 | Total distinct transactions generated by drivers.    |
| Output Packets Expected | 23,676 | Higher count due to Multicast/Broadcast replication. |
| Correctly Matched       | 23,676 | Every expected packet arrived with correct payload.  |
| Routing Errors          | 0      | No packets were misrouted to the wrong port.         |
| Dropped Packets         | 0      | Zero data loss, even during saturation events.       |

## 4.5 Conclusion

The comprehensive verification methodology has demonstrated that the Four-Port Switch design is functionally correct and architecturally robust. The environment proved the design's ability to handle complex routing scenarios, maintain data integrity under maximum load, and recover gracefully from congestion events. The achievement of 100% success rate and 100% functional coverage confirms that the design meets all Stage A and Stage B requirements.